Finding optimum price to maximize the expected gain

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1 Creating the model

Assume we have the unitary cost, unitary price, and the amount of products sold, C, P, Q.

According to the law of demand, the more expensive the product is, the less it is sold, which can be expressed in the next mathematical model:

$$Q = \beta_0 - \beta_1 P + \varepsilon, \quad \beta_0, \beta_1 > 0, \, \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

Or, equivalently,

$$Q \sim \mathcal{N}\Big(\beta_0 - \beta_1 P, \sigma^2\Big).$$

Let be U the gain, it can be expressed as

$$\begin{aligned} U &= Q(P - C) \\ &= (\beta_0 - \beta_1 P + \varepsilon)(P - C) \\ &= \underbrace{(\beta_0 - \beta_1 P)(P - C)}_{\mu_U} + (P - C)\varepsilon. \end{aligned}$$

Or, equivalently,

$$U \sim \mathcal{N}\Big(\mu_U, (P - C)^2 \sigma^2\Big)$$

Note that

$$\mu_U = -\beta_0 C + (\beta_0 + \beta_1 C)P - \beta_1 P^2.$$

We can find $P^\star$ that maximizes the expected gain, μ_U.

$$\frac{\partial \mu_U}{\partial P} = \beta_0 - \beta_1 C - 2\beta_1 P$$

$$\left.\frac{\partial \mu_U}{\partial P}\right|_{P=P^\star} = 0.$$

Thus,

$(P^\star, Q^\star, U^\star)$

$$P^\star = \frac{\beta_0 + \beta_1 C}{2\beta_1}$$

$$Q^\star = \beta_0 - \beta_1 P^\star + \varepsilon$$

$$U^\star = Q^\star (P^\star - C)$$

2 Fitting the model

For instance, assume that we count with the next data:

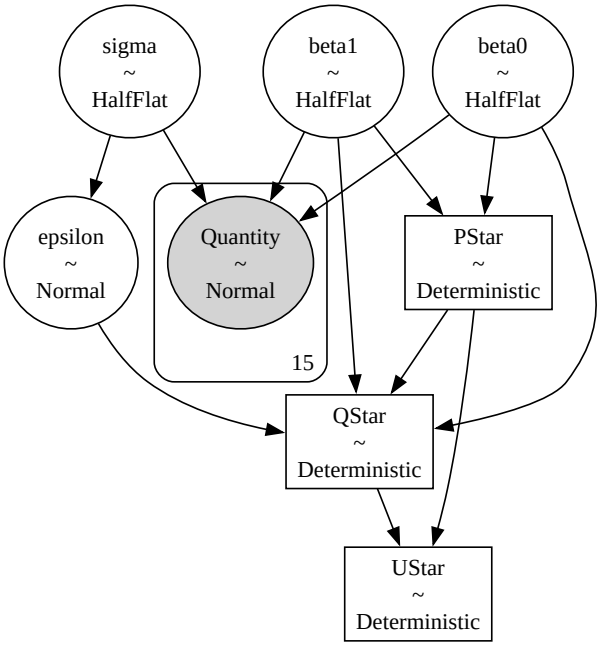
Price	21	19	20	15	18	18	7	16	14	9	11	13	10	10	10
Quantity	3	3	7	6	10	15	16	13	9	15	9	15	12	18	21

With a constant cost $C = 3$.

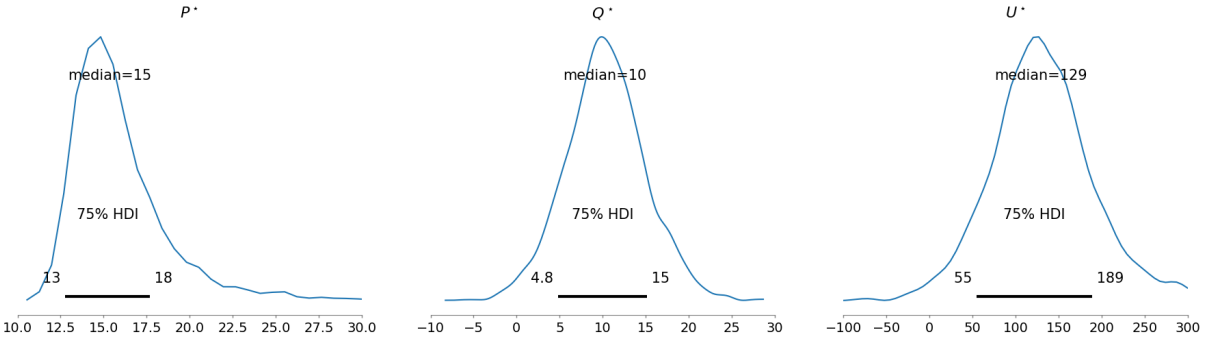
With the next scatter plot between the price and the quantity sold.



Thus, out model can be represented by the next probabilistic graphical model:



Once the model has been fitted, we get the next posterior distributions for $P^\star, Q^\star, U^\star$



3 Confidence and prediction intervals

On the other hand, using

$$Q \sim \mathcal{N}\Big(\mu_Q, \sigma^2\Big), \quad \mu_Q = \beta_0 - \beta_1 P,$$

$$U \sim \mathcal{N}\Big(\mu_U, (P-C)^2\sigma^2\Big), \quad \mu_U = -\beta_0 C + (\beta_0 + \beta_1 C)P - \beta_1 P^2,$$

we can create posterior confidence bands for μ_Q and μ_U, and posterior prediction bands for Q and U.

Taking all these statistics into account, we can represent our results in the next plots, showing the relationship between the price and the quantity sold, and the price and the gain.

